

README for "Appropriate Entrepreneurship? The Rise of China and the Developing World" (Josh Lerner, Junxi Liu, Jacob Moscona, and David Y. Yang)

Journal of Political Economy, manuscript 20250178.

This README follows the template endorsed by the Social Science Data Editors (https://social-science-data-editors.github.io/template_README/), adapted where the mixed public/confidential nature of the package requires it.

Overview

The code in this replication package reproduces every table and figure in the paper and its appendix — 57 exhibits — from raw data, in two stages:

- **Stage A** (`code/generate/`, 25 programs organized in nine dependency-ordered steps) rebuilds every analysis dataset from the raw extracts. The package ships raw data only: `confidential-data-not-for-publication/Analysis/` is empty at delivery and is populated entirely by Stage A.
- **Stage B** (`code/tables/`, `code/figures/`) reproduces every exhibit from the Stage-A datasets with **one self-contained script per exhibit**, writing to `output/tables/` and `output/figures/`.

One command runs everything: `bash code/run_all.sh`. The replicator should expect about 1.5–2 hours for the main analysis (Stage-A steps 01–04 plus Tables 2–8 / Figures 2–4), and roughly 8–10 hours of additional compute for the full appendix, dominated by the Table A8 Monte-Carlo simulation (≈5–6 hours on 7 parallel Stata workers) and two long placebo/robustness builds. A `--skip-heavy` switch skips the two longest builds.

Data fall into two folders, split by redistribution rights:

- `data/raw/` — all **publicly redistributable** raw inputs (World Bank WDI, USPTO PatentsView bulk files, Natural Earth, SimpleMaps/GeoNames gazetteers, UN voting ideal points, Polity5, UN Comtrade extract, and every author-compiled classification file). This folder is part of the public deposit.
- `confidential-data-not-for-publication/` — the **proprietary** inputs (PitchBook extracts; Capital IQ, State Street, Hurun, and Zero2IPO workbooks; and machine-learning outputs derived from PitchBook/PATSTAT text), provided to the JPE Data Editor under the journal's confidential-data exemption, for the sole purpose of the reproducibility checks. This folder is excluded from the public deposit.

The one sanctioned exception to "everything from raw" is a set of nine frozen machine-learning outputs (BERT sector classifications and SBERT text-similarity measures) that are

too computationally intensive to regenerate in the pipeline and are not bit-reproducible (GPU training). The complete original code that produced them, with a file-by-file provenance map, ships in `code/BERT_generation/` (see its `README.md` and `PROVENANCE.md`).

Data Availability and Provenance Statements

- ☒ Confidential data used in this paper and not provided as part of the public replication package will be preserved for 10 years after publication, in accordance with journal policies.

The authors are secondary data users; no original data were collected for this paper through surveys or experiments. All sources, access procedures, and terms are described source-by-source below.

Statement about Rights

- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- ☒ I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package. Appropriate permissions are documented in the `LICENSE.txt` file.

The certification of redistribution rights applies to the public deposit, i.e., everything outside `confidential-data-not-for-publication/`. The contents of that folder are provided to the Data Editor only, under the confidential-data exemption granted for this manuscript, and carry no redistribution rights; see `LICENSE.txt` for the file-level terms.

License for Data

The publicly deposited data are available under their original providers' terms (World Bank WDI: CC-BY 4.0; PatentsView: CC-BY 4.0; Natural Earth: public domain; SimpleMaps World Cities Basic: CC-BY 4.0; OpenDataSoft/GeoNames: CC-BY 4.0; UN voting ideal points: distributed with Bailey, Strezhnev and Voeten's replication materials; Polity5: Center for Systemic Peace terms of use). Author-compiled files in `data/raw/` are licensed under CC-BY 4.0. See `LICENSE.txt` for the full statement, including the non-redistribution terms of the confidential folder.

Summary of Availability

- ☐ All data **are** publicly available.
- ☒ Some data **cannot be made** publicly available.
- ☐ **No data can be made** publicly available.

Summary of Data Availability

Data.Name	Data.Files	Location	Provided	Citation
PitchBoo	pbdeals.dta,	confidential-data-not-	Data	PitchBook

Data.Name	Data.Files	Location	Provided	Citation
k Platform extracts	pbcompanies.dta, pbinvestors.dta, pbdealinvestor.dta, pbmarketmap_v2.csv, later_years_extensions/ (2 files), LP.dta	for- publication/Raw/pitchbook /,.../other_data_resource /	Editor only	Data, Inc. (2024)
ML- derived text measures (from PitchBook/PATSTAT text)	9 files in BERT_prediction_resource/ (+ CSV twins)	confidential-data-not- for- publication/Raw/BERT_prediction_resource/	Data Editor only	derived; see Details
Capital IQ IPO tabulations	3 figure1c_IPO_Capital IQ_*.xlsx workbooks	confidential-data-not- for- publication/Raw/other_data_resource/	Data Editor only	S&P Global (2024)
State Street fund returns	table1_panelC_Copy of VC Returns 2024Q4 v2.xlsx	same folder	Data Editor only	State Street (2024)
Hurun China Rich List tabulation	table1_panelD_Chinese VC.xlsx	same folder	Data Editor only	Hurun (2020)
Zero2IPO China VC series	figureA4_Data from Zero2IPO from HBS China Research Center.xlsx, figureA4_Comparisons of PB and Other Sources.xlsx	same folder	Data Editor only	Zero2IPO (2020)
PitchBook descriptive tabulations (Fig. 1a/1b; VC	figure1ab_Figures 1a and 1b.xlsx, tableA1_VCpc.xlsx, table1_panelA_vc_groups.csv	same folder	Data Editor only	PitchBook Data, Inc. (2024)

Data.Name	Data.Files	Location	Provided	Citation
per capita; VC groups)				
World Bank World Development Indicators	WDI_full.xlsx	data/raw/suitability_resource/	Yes	World Bank (2023)
USPTO PatentsView bulk data	7 g_*.tsv files	data/raw/patent_resource/	Yes	PatentsView (2023–24)
Natural Earth boundaries & populated places	ne_10m_admin_0_countries/ (shapefile set), ne_10m_populated_places.*	data/raw/geocoding_resource/	Yes	Natural Earth (2023)
SimpleMaps World Cities; GeoNames via OpenDataSoft	simplemaps.csv, opendatasoft.csv	data/raw/geocoding_resource/	Yes	SimpleMaps (2023); OpenDataSoft/GeoNames (2023)
UN General Assembly ideal points + Polity5 (combined panel)	un_polity_china.dta (+ .csv)	data/raw/other_data_resource/	Yes	Bailey, Strezhnev & Voeten (2017); CSP (2020)
UN Comtrade extract	Trade_Volume_Export_Import.csv	data/raw/other_data_resource/	Yes	United Nations (2023)
Governm	table1_panelB_fundin	data/raw/other_data_resource/	Yes	Beraja, Peng,

Data.Name	Data.Files	Location	Provided	Citation
ent/private VC funding tabulation	g_sources.csv	rce/		Yang & Yuchtman (2024)
Global Innovation Measures country dataset (author-compiled, supplemented)	CountryDataset.2024019.supplemented.xlsx; tableA1_published_values.csv	workbook: confidential-data-not-for-publication/Raw/other_data_resource/; CSV: data/raw/other_data_resource/	workbook: Data Editor only; CSV: Yes	authors (this package)
Curated global VC series (in-text world VC total)	Global_VC_0525.xlsx	confidential-data-not-for-publication/Raw/other_data_resource/	Data Editor only	NSF NSB (2020); NVCA (2023); others — see Details
Author classification files	hand_collected/ (9 files: 4 classification .dta+ CSV pairs + table6_A16_Tradable or not 2nd.xlsx), country_panel.dta (+ CSV twin), indicators_WDI_MtV_*.xlsx (4), crosswalks	data/raw/	Yes	authors (this package)
EPO PATSTAT (upstream input to one derived file)	not shipped; the derived country-pair counts ship in BERT_prediction_resource/	—	No (derived file: Data Editor only)	EPO (2023)

Details on each Data Source

PitchBook (confidential)

The core data are four extracts from the PitchBook Platform (PitchBook Data, Inc.), covering global venture-capital deals, companies, investors, and the deal–investor link table, plus the analyst "market map" taxonomy (pbmarketmap_v2.csv: companyid × marketmap|segment|subsegment with company descriptions), a 2022–24 refresh (later_years_extension/), and a fund–limited-partner extract (LP.dta). Format: Stata .dta / .csv; the variable lists for all PitchBook files are reproduced in full in the Data Appendix of this README. The extracts were downloaded by the authors through Harvard Business School's institutional PitchBook subscription (2023 baseline; the 2022–24 refresh pulled in 2025). PitchBook data are proprietary and may not be redistributed; researchers can obtain access through an institutional or commercial PitchBook license (<https://pitchbook.com>; academic licensing is negotiated with PitchBook's institutional sales — there is no posted price). The files are provided to the JPE Data Editor under the journal's confidential-data exemption, to be destroyed after the reproducibility checks; they are excluded from the public deposit. The authors will assist with reasonable replication attempts for ten years following publication.

Machine-learning outputs derived from PitchBook and PATSTAT text (confidential)

confidential-data-not-for-publication/Raw/BERT_prediction_resource/ holds the nine frozen outputs of the paper's three expensive ML layers: (1) BERT market-map sector classification of every PitchBook company description (company_level_predictions_v2.dta); (2) SBERT business-description similarity to Chinese and U.S. firms (similarity_compiled.csv, similarity_compiled_western.csv, similarity_at_sector_x_country_x_year.dta); (3) patent-text sector labels and similarity (patent_sector_labels_{grantyear,new_grantyear,new100k}.dta, patent_countrysector_similarity_2000_2013_100k_new_china.dta, worldwide_pairs_filtered_no_EPO_countries.dta — the last derived from EPO PATSTAT family abstracts). Because they embed PitchBook text (and, for the last file, PATSTAT-licensed inputs), they are confidential. Every file's producing code, model, hyperparameters, inputs, and compute cost are documented in code/BERT_generation/ (original author code plus PROVENANCE.md); regenerating them is optional and not bit-reproducible (GPU training). Every .dta file ships with a CSV twin (see Dataset list).

Capital IQ (confidential)

Figure 1 Panel C compares IPO counts from S&P Global's Capital IQ platform with PitchBook-based counts. Three author-built tabulation workbooks (figure1c_IPO_CapitalIQ_to_PB_{EM,US,nonEM_nonUS}*.xlsx) contain the Capital IQ screens and the mapping to the paper's country groups. The non-EM/non-US workbook additionally embeds the authors' manual, research-based corrections of the Capital IQ output (the case-by-case adjustments described in the paper's appendix); these hand edits are part of the shipped workbook and are not mechanically reproducible. Capital IQ

is proprietary (<https://www.capitaliq.spglobal.com>; institutional subscription); the workbooks are provided to the Data Editor only.

State Street (confidential)

Table 1 Panel C reports pooled internal rates of return and Kaplan–Schoar (2005) public-market-equivalents for venture-capital funds by regional focus, over 2007Q3–2019Q4. State Street Corporation, one of the world's largest custodians (roughly \$49 trillion in assets under custody and/or administration as of June 30, 2025), computes these returns in its custodial role, maintaining a comprehensive record of all investor–GP cash flows (net of management fees and carried interest) across VC, private equity, real estate, hedge funds, securitizations, and other alternative assets. The underlying universe spans 142 large asset owners with VC and PE exposure, collectively over 120,000 investments, and 4,008 private-capital funds (vintages 1988–2024) covering VC (early-stage, balanced, and late-stage), buyout, and private-debt strategies (distressed, mezzanine, and special lending); State Street classifies funds by type from documents provided by the fund managers and asset owners. Panel C tabulates only the venture-capital funds, broken out by the region each specialized in; funds without a clear geographic focus, or spanning multiple regions, are excluded from the tabulation. The data were provided by Nan Zhang (Head of Product Development and Managing Director, State Street Data Intelligence) and pulled from the database on June 24, 2025; see Lerner, Mao, Schoar and Zhang (2022) for a detailed discussion and validation of these data. The provider summary workbook (table1_panelC_...xlsx) is not redistributable and has no public distribution channel; access is negotiated with State Street's Data Intelligence group. Provided to the Data Editor only.

Hurun China Rich List (confidential)

Table 1 Panel D's founder counts are hand-tabulated from the Hurun China Rich List 2020 (Hurun Research Institute, <https://www.hurun.net>), matched by the authors to PitchBook company records to establish VC backing. The tabulation workbook (table1_panelD_Chinese VC.xlsx) is provided to the Data Editor only.

Zero2IPO (confidential)

Figure A4 cross-validates PitchBook's China venture series against Zero2IPO (清科) data obtained through the Harvard Business School China Research Center. Zero2IPO data are licensed (<https://www.pedata.cn>); the two comparison workbooks are provided to the Data Editor only. RMB values are converted to USD in code.

World Bank World Development Indicators (public; provided)

The appropriateness/suitability measure is built from the World Development Indicators. data/raw/suitability_resource/WDI_full.xlsx is the pinned bulk extract used by the paper (all countries × indicators × 1960–2021, with WDI series codes). WDI data are CC-BY 4.0 and can be re-downloaded at <https://databank.worldbank.org/source/world-development-indicators> (choose "All countries", "All series", "All years", Excel) — note that

the World Bank revises the database continuously (including retroactive revisions and indicator retirements), so an exact re-download of this vintage is not possible; the pinned extract ships instead. The four author-built indicator→sector assignment maps ship in the same folder (indicators_WDI_MtV_*.xlsx; the corrected map is the one used by the paper, the other three feed the Table A4 assignment-robustness comparison). The assignment of WDI indicators to the fifteen market-map sectors is likewise the authors' own manual classification (the gpt_assigned variant, which enters only the Table A4 comparison, was instead produced with a large language model).

USPTO PatentsView (public; provided)

The entire patent layer is rebuilt from seven raw PatentsView bulk TSVs (g_patent, g_us_patent_citation, g_cpc_current, g_application, g_assignee_disambiguated, g_location_disambiguated, g_inventor_disambiguated; 21 GB total). PatentsView (<https://patentsview.org>, downloads at <https://patentsview.org/download/data-download-tables>) is free, CC-BY 4.0, no registration required. PatentsView publishes quarterly releases and does not archive superseded vintages, so the package ships the authors' pinned copies (the CPC table is the February 2023 release; the citation table is the November 2023 release; the remaining tables were downloaded in 2023–24). A newer release would shift disambiguation and therefore downstream counts; the shipped vintage reproduces the paper.

Natural Earth; SimpleMaps; OpenDataSoft/GeoNames (public; provided)

Company and patent geolocation uses: Natural Earth 1:10m Admin-0 country boundaries and Populated Places (public domain, <https://www.naturalearthdata.com>); the SimpleMaps World Cities Database, Basic edition (CC-BY 4.0, <https://simplemaps.com/data/world-cities>); and the "All cities with a population > 1000" GeoNames extract republished by OpenDataSoft (CC-BY 4.0, <https://public.opendatasoft.com/explore/dataset/geonames-all-cities-with-a-population-1000/>). All are shipped in data/raw/geocoding_resource/, along with the authors' country/state name crosswalk (crosswalk.xlsx).

UN General Assembly ideal points and Polity5 (public; provided)

The political-proximity control uses dynamic UN-voting ideal points from Bailey, Strezhnev and Voeten (2017), obtained from the paper's public replication data (<https://doi.org/10.7910/DVN/LEJUQZ>), and regime scores from the Polity5 Annual Time-Series 1946–2018 (Center for Systemic Peace, free download at <https://www.systemicpeace.org/inscrdata.html>). The two are shipped combined in one country–year panel assembled for the authors' earlier work (data/raw/other_data_resource/un_polity_china.dta, with CSV twin), carrying each country's values alongside China's. The distance-to-China controls are computed in code (code/generate/03b_generate_controls.py): $un_distance = |un_ideal_point - china_un_ideal_point|$ and $polity_distance = |polity_score -$

china_polity_score|. Coverage note: Polity reaches back before the UN series (China's ideal points begin 1971), so early years have missing UN values.

UN Comtrade (public; provided)

The China trade-share control uses a UN Comtrade extract (Trade_Volume_Export_Import.csv: annual total-trade flows with China, in the standard Comtrade bulk column layout). Comtrade data are free at <https://comtradeplus.un.org> (registration optional for small queries). The extract ships in the package.

Government vs. private Chinese VC (public; provided)

Table 1 Panel B's classification of Chinese VC funding by government/private source is tabulated from the data of Beraja, Peng, Yang and Yuchtman (2024). The tabulation (table1_panelB_funding_sources.csv, 3 columns) ships in the package; the underlying data are available with that paper.

Global Innovation Measures country dataset (author-compiled; supplemented; confidential)

Table A1 and Figure A1 draw on an author-compiled country-year workbook (CountryDataset.2024019.supplemented.xlsx; confidential folder) combining VC activity, GDP per capita, R&D spending, scientific publications, and U.S. patents by country, 1969–2021. The workbook is the authors' own compilation, assembled by hand rather than by any program in the package: the underlying series were collected from the sources described next and cleaned and harmonized directly in the workbook. The VC columns are consolidated from the authors' PitchBook Platform pulls (deal counts and amounts by country and year, from the same platform extracts as the core data). The publications series (1996 onward) is taken directly from NSF Science and Engineering Indicators country counts; the only change is renaming entries to a standard country-naming convention used across the measures (the 1973–1995 counts are the Web of Science supplement described below). The R&D series splices three sources — OECD (1981–1995), World Bank (1996–2014), and UNESCO (2015–2021) — with judgmental choices between overlapping vintages; short interior gaps are linearly interpolated and series ends are carried forward, but values are never back-filled before a country's first observation. Because of the judgment involved, the package ships the compiled workbook (all cleaning sheets included, documenting every source series and imputation) plus the exhibit code, rather than a from-scratch reconstruction. CountryDataset.2024019.supplemented.xlsx is the authors' compiled Global Innovation Measures workbook (CountryDataset.2024019) with three documented supplements added for replication: (1) pre-1996 publication counts inserted from a Science Citation Index (Web of Science) country-count extract, 1973–1995 (compiled by Kathleen Ryan, Baker Research & Data Services, accessed 6/5/2023) — the reason the workbook is confidential; (2) explicit zeros for three pre-coverage VC cells used by Table A1 (Israel 1969, Singapore 1979, Chile 1993; no recorded deals); (3) two imputed R&D levels carried from the authors' original hand tabulation, South Korea 1988 (2.0% of GDP) and Chile 1993 (0.78% of GDP) — assumptions, not

source data. With these supplements, Table A1 is derived entirely from this workbook plus tableA1_VCpc.xlsx, and the per-cell source of every Table A1 entry is recorded in output/intermediate/tableA1_source_audit.csv. tableA1_published_values.csv is an author transcription of Table A1 as printed in the manuscript; it no longer supplies any table value – table_A1.py takes from it only the country list, the emergence years, and the print-N/A gating for the VC-per-capita column, and recomputes every entry from the workbook; tableA1_VCpc.xlsx (VC per capita, PitchBook-derived) is in the confidential folder.

Curated global venture-investment series (author-compiled; confidential)

Global VC_0525.xlsx (confidential folder) is the authors' curated compilation of published venture-investment tabulations: annual VC invested, in current US\$ billions, for the US, China, Western Europe, Canada, Japan, Australia, a developing-country residual, and the world, 2001–2022. The 2001–2018 base is the National Science Board's Science and Engineering Indicators 2020, Table S8-62 (itself PitchBook-sourced); the 2019–22 values are extended region by region from the NVCA Yearbook 2023 (US and world), the authors' PitchBook analysis and PitchBook-sourced press reporting (China), and Invest Europe, CVCA, Initial Enterprise, and Cut Through Venture (other regions), converted at IRS yearly-average exchange rates; every source and adjustment is noted in the workbook itself. No exhibit or program reads this file: it is the quoted source for the manuscript's 2021 world venture-investment total (opening Background paragraph); all other in-text numbers are cells of the exhibits or immediate arithmetic on them. Provided to the Data Editor only.

Hand-collected classification files (author-compiled; provided)

data/raw/other_data_resource/hand_collected/ contains nine author files — four classification .dta files with CSV twins plus one .xlsx (full variable lists in the Data Appendix): OECD membership and join years (OECD_list.dta; the paper's EM definition is non-OECD as of 1980); sector policy-constraint codings hand-linked from public policy documents (EM_policy_constraints.dta, policy_constrained_sectors.dta); critical/state-supported technology flags mapping "Made in China 2025", critical-supply-chain, and U.S. DoD critical-technology lists to PitchBook subsegments (critical_subseg.dta); and the tradable-sector classification behind Table 6/A16 and Figure A16 (table6_A16 Tradable or not 2nd.xlsx). The Table A1 published-values CSV, the Panel B funding-shares tabulation documented above, and the country panel of bilateral gravity/distance measures relative to China (country_panel.dta + CSV twin, assembled from standard gravity sources; see the paper's Data Appendix) sit at the other_data_resource/ folder root; the supplemented Global Innovation Measures workbook is in the confidential folder (see above). The deal-type → category classification for the non-VC analyses is hardcoded in code/generate/05c_generate_regression_alltype_by_cat.do (formerly a spreadsheet).

EPO PATSTAT (upstream; not shipped)

Table A5's jointly-filed patent-family counts derive from the EPO Worldwide Patent Statistical Database (PATSTAT Global; licensed product of the European Patent Office, <https://www.epo.org/en/searching-for-patents/business/patstat> — annual subscription, on the order of €1,250/year for the Global edition). PATSTAT itself is not shipped; only the derived country-pair × sector family counts (worldwide_pairs_filtered_no_EPO_countries.dta, confidential folder, with CSV twin) enter the pipeline. The complete producing code is code/BERT_generation/3_patent_ml/patent_family_exercise.ipynb.

Dataset list

All raw inputs, by folder. "Twin" = a non-proprietary .csv copy of the same data ships alongside the .dta (exported with pandas; identical row/column counts verified). Shapefiles, .tsv, .csv, and .xlsx are open/standard formats readable by free software.

Public — data/raw/ (~21.3 GB):

Data file	Source	Notes	Provided
suitability_resource/WDI_full.xlsx	World Bank WDI	pinned bulk extract; input to step 02	Yes
suitability_resource/indicators_WDI_MtV_corrected.xlsx	authors	indicator→sector map used by the paper	Yes
suitability_resource/indicators_WDI_MtV_{from_existing_no_trade2, from_scratch_allassigned, gpt_assigned}.xlsx	authors	alternative maps; Table A4 robustness	Yes
suitability_resource/country_crosswalk.dta (+ twin)	authors	ISO-2 ↔ country-name crosswalk	Yes
patent_resource/g_patent.tsv (6.1 GB)	PatentsView	patent universe, grant dates, abstracts	Yes
patent_resource/g_us_patent_citation.tsv (8.9 GB)	PatentsView	citation pairs (Nov-2023)	Yes

Data file	Source	Notes	Provided
patent_resource/g_cpc_current.tsv (2.7 GB)	PatentsView	release) CPC classes (Feb-2023 release)	Yes
patent_resource/g_application.tsv (379 MB)	PatentsView	filing dates	Yes
patent_resource/g_assignee_disambiguated.tsv (946 MB)	PatentsView	assignees	Yes
patent_resource/g_location_disambiguated.tsv (8 MB)	PatentsView	assignee locations	Yes
patent_resource/g_inventor_disambiguated.tsv (1.9 GB)	PatentsView	inventors	Yes
geocoding_resource/ne_10m_admin_0_countries/ (shapefile set)	Natural Earth	country polygons	Yes
geocoding_resource/ne_10m_populated_places.{shp,dbf,shx,prj,cpg}	Natural Earth	populated places; converted by step 03a	Yes
geocoding_resource/simplemaps.csv	SimpleMaps Basic	world cities gazetteer	Yes
geocoding_resource/opendatasoft.csv	GeoNames/OpenDataSoft	cities >1000 population	Yes
geocoding_resource/crosswalk.xlsx	authors	country/US-state name harmonization	Yes
other_data_resource/un_polity_china.dta (+ twin)	Bailey et al. (2017); CSP Polity5	combined country-year panel	Yes
other_data_resource/Trade_Volume_Export_Import.csv	UN Comtrade	China trade flows	Yes
other_data_resource/hand_collected/OECD_list.dta (+ twin)	authors	OECD membership; EM definition	Yes
other_data_resource/country_panel.dta (+ twin)	authors (gravity sources)	bilateral controls rel. to China	Yes

Data file	Source	Notes	Provided
other_data_resource/hand_collected/critical_subseg.dta (+ twin)	authors	MIC2025/CST/DoD × subsegment flags	Yes
other_data_resource/hand_collected/EM_policy_constraints.dta (+ twin)	authors	policy × country constraint panel	Yes
other_data_resource/hand_collected/policy_constrained_sectors.dta (+ twin)	authors	policy × subsegment map	Yes
other_data_resource/tableA1_published_values.csv	authors	Table A1 row list, emergence years, and print-N/A gating (transcribed from the manuscript; supplies no values)	Yes
other_data_resource/table1_panelB_funding_sources.csv	Beraja et al. (2024)	Panel B tabulation	Yes
other_data_resource/hand_collected/table6_A16 Tradable or not 2nd.xlsx	authors	tradable-sector classification	Yes

Confidential — confidential-data-not-for-publication/Raw/ (~24 GB; Data Editor only):

Data file	Source	Notes	Provided
pitchbook/pbdeals.dta (9.1 GB, 95 vars) (+ twin)	Pitch Book	all deals	Editor only
pitchbook/pbcompanies.dta (4.8 GB, 94 vars) (+ twin)	Pitch Book	all companies incl. descriptions	Editor only

Data file	Source	Notes	Provided
pitchbook/pbinvestors.dta (2.3 GB, 94 vars) (+ twin)	Pitch Book	all investors	Editor only
pitchbook/pbdealinvestor.dta (0.9 GB, 11 vars) (+ twin)	Pitch Book	deal-investor links	Editor only
pitchbook/pbmarketmap_v2.csv (13 MB)	Pitch Book	analyst market map + descriptions	Editor only
pitchbook/later_years_extension/All_VC_Comp_Deal_Investor_2022-2024.csv (4.9 GB)	Pitch Book	2022-24 refresh	Editor only
pitchbook/later_years_extension/pbmarketmap_22_24.csv (77 MB)	Pitch Book	2022-24 market map	Editor only
other_data_resource/LP.dta (+ twin)	Pitch Book	fund × LP-country extract (Table A9 input)	Editor only
BERT_prediction_resource/company_level_predictions_v2.dta (1.1 GB) (+ twin)	derived (BERT on Pitch Book text)	company sector predictions; producing code in code/BERT_generation/	Editor only
BERT_prediction_resource/similarity_compiled.csv, similarity_compiled_western.csv	derived (SBE RT)	company-level similarity to CN/US firms	Editor only
BERT_prediction_resource/similarity_at_sector_x_country_x_year.dta (+ twin)	derived (SBE RT)	sector×country×year collapse	Editor only
BERT_prediction_resource/patent_sector_labels_{grantyear,new_grantyear,new100k}.dta (+ twins)	derived (BERT on patent)	patent→sector labels, two prediction runs	Editor only

Data file	Source	Notes	Provided
	t text)		
BERT_prediction_resource/patent_countrysector_similarity_2000_2013_100k_new_china.dta (+ twin)	derived (SBE RT)	patent-text similarity (Table A5)	Editor only
BERT_prediction_resource/worldwide_pairs_filtered_no_EPO_countries.dta (+ twin)	derived (BERT + PATS TAT)	family country-pair counts (Table A5)	Editor only
other_data_resource/ — 7 vendor workbooks + 3 author-corrected/classified files + the supplemented Global Innovation Measures workbook (CountryDataset.2024019.supplemented.xlsx) + the curated global VC series (Global VC_0525.xlsx), all at the folder root	Pitch Book / Capital IQ / State Street / Hutton / Zero2 IPO / authors	descriptive-exhibit sources (Table 1, Figs. 1, A4, Table A1 VC p.c.; Table A1/Figure A1 innovation measures)	Editor only

Every .dta raw input ships with a CSV twin (same basename, .csv extension, in the same folder), so no data in the package is available only in a proprietary format. The twins were exported with pandas and verified to have identical row and column counts; Stata dates appear in ISO format, and text fields are standard-quoted. The pipeline itself reads the .dta files (which additionally carry the original variable labels).

confidential-data-not-for-publication/Analysis/ is **empty at delivery** and is populated by Stage A (two core files at its root — analysis_v2.dta, regression_corrected_120623.dta — and everything else under intermediate_and_other_data/).

Computational requirements

Software Requirements

- ☒ The replication package contains programs to install all dependencies: `code/auxiliary/install.do` (Stata) and `requirements.txt` (Python).
- **Stata:** Stata 19 SE or later (the generators set version 19; the code was last run under StataNow 19.5 SE). Tested on macOS/Apple Silicon and Windows 11/x64 with identical output values. Community packages (installed once by `code/auxiliary/install.do`): `ftools`, `reghdfe`, `require`, `estout`, `ivreghdfe`, `ivreg2`, `ranktest`, `geonear`, `shp2dta`, `nrow` (from SSC), `binscatter2` (from GitHub). The graph scheme `scheme-eop.scheme` ships in `code/auxiliary/` and is found via the `adopath` set by `paths.do`.
- **Python** ≥ 3.11 (last run with 3.11) with the pinned packages in `requirements.txt`: `pandas==2.2.3`, `numpy>=1.26,<3`, `matplotlib==3.10.0`, `seaborn==0.13.2`, `geopandas==0.14.4`, `scipy>=1.11`, `openpyxl>=3.1`, `pycountry>=23.12`, `Unidecode>=1.3`. Install with `pip install -r requirements.txt`.
- **Shell:** the two drivers (`run_all.sh`, `generate_all.sh`) are bash scripts (macOS/Linux native; on Windows use Git Bash or WSL, or run the individual programs in the documented order).
- The optional ML-layer rerun (`code/BERT_generation/`) has its own `requirements.txt` (PyTorch, transformers, sentence-transformers) and GPU requirements; it is **not** part of the standard reproduction.

Controlled Randomness

- ☒ Random seed is set at lines 97 and 162 of `code/generate/09_generate_simulated_deals.do` (the Table A8 Monte-Carlo: the seed for each simulation draw is the draw index, set immediately before each sample call, so the 500 draws are reproducible individually and in parallel).
- ☒ Random seed is set at lines 111 and 147 of `code/figures/figure_3.do` (the 500-draw placebo loops behind Figure 3, Panels B and C; seed = draw index).
- ☒ Random seed is set at line 241 of `code/generate/02_generate_suitability.py` (the appropriateness-score placebo component; seeded per iteration).
- ☒ Sort order is pinned wherever ties could otherwise be machine-dependent: set `sortseed` at lines 62, 84, and 171 of `code/generate/03c_generate_regression_auxiliary.do`, plus unique-key sorts throughout the generators.
- No other program uses random number generation. Outputs are reproducible across reruns and across operating systems; the authors verified macOS/arm64 vs. Windows/x64 agreement to at most $1.2e-07$ in any regression statistic, with all displayed values identical.

Memory, Runtime, Storage Requirements

Summary time to reproduce

Approximate time needed to reproduce the analyses on a standard (2026) desktop machine:

- ☐ <10 minutes
- ☐ 10-60 minutes
- ☒ 1-2 hours — the main analysis: Stage-A steps 01–04 plus Tables 2–8 and Figures 2–4
- ☐ 2-8 hours
- ☒ 8-24 hours — the complete package including all appendix exhibits, with step 09 run on 7 parallel workers (roughly 2 days if everything is run sequentially on a single worker)
- ☐ 1-3 days
- ☐ 3-14 days
- ☐
14 days

Summary of required storage space

Approximate storage space needed:

- ☐ < 25 MBytes
- ☐ 25 MB - 250 MB
- ☐ 250 MB - 2 GB
- ☐ 2 GB - 25 GB
- ☒ 25 GB - 250 GB — the package is ~45 GB as delivered; Stage A writes roughly another 70–80 GB of generated data (largest single generated file ≈9 GB); plan for ~120 GB free disk in addition to the delivered package.
- ☐
250 GB

Computational Details

The package was last run in full on an **Apple MacBook Pro (M3 Max, 14 cores, 64 GB RAM), macOS 26**, with StataNow 19.5 SE and Python 3.11; values were cross-checked on a Windows 11 x64 workstation. 64 GB RAM is recommended: the keystone build sets set maxvar 30000 and the largest single input is a 9.1 GB PitchBook extract (loaded whole by Stata). 32 GB machines can complete steps 01–07 but should not run more than 3–4 parallel workers in step 09.

Measured wall-clock times (machine above):

Stage A step	Content	Time
01	deal/company base (analysis_v2)	≈ 4 min
02	appropriateness-score family	≈ 4.5 min
03a–03c	populated places; controls; regression auxiliaries	≈ 2.5 min
04	keystone regression panel	≈ 8.5 min
05a–05i	robustness/extension inputs	≈ 9 min
06a–06c	patent layer (streams the 8.9 GB citation table)	≈ 13 min
07a–07d	geolocation + city panel	≈ 10 min
08a	Table A13 variant regressions	≈ 35 min
08b	figure-A11 placebo stacks	≈ 50 min
09	Table A8 Monte-Carlo (500 draws × 2 designs)	≈ 5–6 h on 7 workers

Stage B: seconds to ~1 minute per exhibit, except figure_3.do (~55 minutes: two 500-draw placebo loops run inside the exhibit program).

Description of programs/code

- code/auxiliary/paths.do / paths.py define every path relative to the package root; **no machine-specific path exists anywhere in the code**. Every program locates the root itself by walking up the directory tree to code/auxiliary/paths.do, so programs can be run from the package root or any folder inside it (or set the environment variable REPLICATION_ROOT / JPE_GENERATE_ROOT explicitly). Public raw data are read from data/raw/; confidential raw data from confidential-data-not-for-publication/Raw/; generated data are written under confidential-data-not-for-publication/Analysis/ (they are PitchBook-derived and therefore confidential). The path setup recreates Analysis/ and the output/ tree if absent.
- code/auxiliary/install.do installs the required Stata community packages (run once). code/auxiliary/scheme-eop.scheme is the graph scheme.
- code/generate/ — **Stage A**: 25 programs in nine dependency-ordered steps (driver: generate_all.sh, which accepts --from NN --to NN step ranges and --skip-heavy):
 - 01** builds the deal/company base analysis_v2.dta from the PitchBook extracts + BERT sector predictions + SBERT similarity + OECD list.
 - 02** builds the appropriateness (suitability) score family from the WDI extract and the indicator→sector maps (paper measure plus all-country, alternative-assignment, and placebo variants).
 - 03a** converts the Natural Earth populated-places shapefile to .dta; **03b** builds the country-year controls (internet penetration, China trade/FDI shares, UN/Polity distances); **03c** builds regression auxiliaries (investor-origin flags, LP shares from LP.dta, the shock-year file, serial-founder inputs).

- **04** assembles the keystone regression panel `regression_corrected_120623.dta` (sector×country×year). After 01–04, Tables 2–8 and Figures 2–4 reproduce. The keystone file is **never modified by any later step**.
- **05a–05i** build robustness/extension inputs: post-regression inputs (05a), the GDP-suitability component (05b), deal-type-by-category and cluster panels (05c, 05d), pre-period deals (05e), western-split similarity (05f), the patent/PATSTAT validation panel for Table A5 (05g), and the 2022–24 extension (05h, 05i).
- **06a** streams the raw PatentsView citation table into located-citation files; **06b** builds the patent sector layers; **06c** builds a patent-augmented **copy** of the keystone (`regression_corrected_120623_patents.dta`, read only by Table A6).
- **07a–07d** build CPC percentile flags, patent geolocation, company geolocation, and the city-level panel (`analysis_city`).
- **08a** re-estimates the Table A13 specification variants; **08b** builds the figure-A11 dropped-indicator placebo stacks.
- **09** runs the Table A8 Monte-Carlo simulation (500 draws × 2 designs; parallel driver `09_generate_simulated_deals.sh <workers>`).
- `code/tables/`, `code/figures/` — **Stage B**: one self-contained program per exhibit (`table_1.py ... table_A30.do`, `figure_1.py ... figure_A17.do`; `.py` for the workbook-based descriptive exhibits, `.do` otherwise). Each program reads only Stage-A datasets and/or raw workbooks and writes its exhibit to `output/`.
- `code/BERT_generation/` — the original author code for the frozen ML layer, with a file-by-file provenance map (`PROVENANCE.md`): the BERT company classifier, the SBERT similarity pipeline, the patent classifiers (including the SLURM setup used for full-corpus inference), and a verified reconstruction program for the three `patent_sector_labels_*` files. Rerunning is optional and not bit-reproducible (GPU training); the frozen outputs are the paper's numbers.
- `code/run_all.sh` — master driver: Stage A, then every Stage-B exhibit in order (flags: `--skip-heavy`, `--stage-a-only`, `--stage-b-only`).

One dataset-level note: the patent layer is rebuilt **entirely** from the raw PatentsView TSVs — located citations, citation counts, CPC percentile flags, geolocation — with zero frozen patent inputs except the BERT/SBERT text measures. The shipped citation table is the only surviving vintage; it postdates the one used for the paper's `num_citations` variable, which shifts ~80k cells of `analysis_city` in columns no regression uses — Tables A29/A30 and Figure A17 are unaffected (verified byte-identical from either vintage).

License for Code

The code is licensed under a BSD-3-Clause license. See `LICENSE.txt` for details.

Instructions to Replicators

1. Unzip the package (and the confidential archive provided separately) so that data/, confidential-data-not-for-publication/, code/, and output/ sit side by side under one root. confidential-data-not-for-publication/Analysis/ is empty by design.
2. Install the Python environment: `pip install -r requirements.txt` (Python ≥ 3.11 ; a fresh virtual environment is recommended).
3. Install the Stata community packages once: `/path/to/stata-se -b do code/auxiliary/install.do`.

4. **One command for everything:**

```
bash code/run_all.sh                # full reproduction (~8–10 h; see r
untimes)
bash code/run_all.sh --skip-heavy    # skips steps 08b + 09 (Figure A11
and
                                     # Table A8 are then not produced)
```

Stata is auto-detected (stata-se/stata-mp on PATH or the standard macOS/Windows install locations); override with the environment variable STATA_CMD. Python is auto-detected; override with PYTHON_CMD.

5. Or run the stages separately:

```
bash code/generate/generate_all.sh          # Stage A, all st
eps
bash code/generate/generate_all.sh --from 01 --to 04 # any contiguous
range
bash code/generate/generate_all.sh --skip-heavy      # all except 08b
+ 09
bash code/generate/09_generate_simulated_deals.sh 7  # step 09 on 7 wo
rkers
```

After steps 01–04 (~20 minutes) the keystone panel exists and Tables 2–8 / Figures 2–4 can be reproduced; each later step unlocks the exhibits listed in the table below. Do not exceed ~7 parallel workers for step 09 on a 64 GB machine (each worker loads the full deal panel).

6. Stage B, one program per exhibit, in any order (or `bash code/run_all.sh --stage-b-only`):

```
/path/to/stata-se -b do code/tables/table_2.do
python3 code/figures/figure_1.py
```

(Windows: `StataSE-64.exe -e do code\tables\table_2.do`.)

7. Outputs land in output/tables/ (.csv, one file per table, plus per-panel files where the exhibit has panels) and output/figures/ (.pdf); Stata logs in output/logs/. Some exhibit programs also write small by-products to output/intermediate/: the exact numeric series plotted in the figures (e.g., `figure1ab_real_vc_2011usd.csv`) and per-panel audit files for the workbook-based tables (e.g., `tableA1_source_audit.csv`). These exist only so every plotted or tabulated value

can be inspected as plain text — no program reads them back, and the pipeline does not depend on this folder.

Details on various programs

Stage-A step → exhibits it unlocks (all steps are driven by `generate_all.sh`; every step assumes the previous ones have run):

After step	These exhibits can be reproduced
01–04	Tables 2, 3, 4, 5, 6, 7, 8; Figures 2, 3, 4, A2, A3, A5, A6, A7, A8, A9, A12, A13, A16; Tables A2, A4, A7, A9, A14, A15, A16, A17, A18, A20, A22, A24, A25
05	Tables A5, A10, A11, A12, A21, A23, A26, A27, A28; Figures A10, A14, A15
06	Table A6
07	Tables A29, A30; Figure A17
08	Table A13; Figure A11
09	Table A8
(raw workbooks)	Table 1, Table A1, Figure 1, Figure A1, Figure A4 (Table 1 also needs step 01)

No exhibit is reproducible from the public folder alone: Table A1 and Figure A1 read the supplemented Global Innovation Measures workbook (confidential; it embeds Web of Science–derived publication counts), and every other exhibit requires the confidential PitchBook layer upstream.

List of tables and programs

The provided code reproduces:

- ☐ All numbers provided in text in the paper
- ☒ All tables and figures in the paper
- ☐ Selected tables and figures in the paper, as explained and justified below.

(The numbers quoted in the text are cells of the exhibits or immediate arithmetic on them, with one documented exception: the 2021 world venture-investment total in the opening Background paragraph, which is quoted from the authors' curated compilation Global VC_0525.xlsx in the confidential folder (see Details on each Data Source). There is no separate in-text-numbers program.)

Every exhibit is produced by the identically-numbered program; outputs carry the same number (code/tables/table_A12.do → output/tables/tableA12.csv). Tables A3 and A19 are textual/illustrative lists (example indicators; policy lists) with no computed content and are not part of the code package.

Figure/Table #	Program	Output file(s)	Note
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Figure/Table #	Program	Output file(s)	Note
Table 1	code/tables/table_1.py	table1.csv (+ per-panel audit files)	panels from source workbooks + step-01 aggregates
Table 2	code/tables/table_2.do	table2.csv	
Table 3	code/tables/table_3.do	table3.csv (+ panels A/B)	
Table 4	code/tables/table_4.do	table4.csv	
Table 5	code/tables/table_5.do	table5.csv	
Table 6	code/tables/table_6.do	table6.csv	uses tradable classification
Table 7	code/tables/table_7.do	table7.csv (+ panels A/B)	
Table 8	code/tables/table_8.do	table8.csv (+ panels A/B)	
Table A1	code/tables/table_A1.py	tableA1.csv (+ source audit)	
Table A2	code/tables/table_A2.do	tableA2.csv	
Table A4	code/tables/table_A4.do	tableA4.csv	needs step 06
Table A5	code/tables/table_A5.do	tableA5.csv	
Table A6	code/tables/table_A6.do	tableA6.csv	
Table A7	code/tables/table_A7.do	tableA7.csv	
Table A8	code/tables/table_A8.do	tableA8.csv	
Table A9	code/tables/table_A9.do	tableA9.csv	needs step 09
Table A10	code/tables/table_A10.do	tableA10.csv	
Table A11	code/tables/table_A11.do	tableA11.csv	
Table A12	code/tables/table_A12.do	tableA12.csv	
Table A13	code/tables/table_A13.do	tableA13.csv	
Table A14	code/tables/table_A14.do	tableA14.csv (+ panels)	2022–24 extension (Panels C/D)
Table A15	code/tables/table_A15.do	tableA15.csv	
Table A16	code/tables/table_A16.do	tableA16.csv	
Table A17	code/tables/table_A17.do	tableA17.csv	
Table A18	code/tables/table_A18.do	tableA18.csv	
Table A20	code/tables/table_A20.do	tableA20.csv	

Figure/Table #	Program	Output file(s)	Note
Table A21	code/tables/table_A21.do	tableA21.csv	
Table A22	code/tables/table_A22.do	tableA22.csv	
Table A23	code/tables/table_A23.do	tableA23.csv	
Table A24	code/tables/table_A24.do	tableA24.csv	
Table A25	code/tables/table_A25.do	tableA25.csv	
Table A26	code/tables/table_A26.do	tableA26.csv	
Table A27	code/tables/table_A27.do	tableA27.csv	
Table A28	code/tables/table_A28.do	tableA28.csv	
Table A29	code/tables/table_A29.do	tableA29.csv	needs step 07
Table A30	code/tables/table_A30.do	tableA30.csv	needs step 07
Figure 1	code/figures/figure_1.py	figure1a.pdf, figure1b.pdf, figure1c.pdf	
Figure 2	code/figures/figure_2.do	figure2a.pdf, figure2b.pdf	
Figure 3	code/figures/figure_3.do	figure3a.pdf, figure3b.pdf, figure3c.pdf	~55 min
Figure 4	code/figures/figure_4.do	figure4a.pdf, figure4b.pdf	
Figure A1	code/figures/figure_A1.py	figureA1a.pdf, figureA1b.pdf	input is the supplemented workbook (confidential)
Figure A2	code/figures/figure_A2.do	figureA2.pdf	
Figure A3	code/figures/figure_A3.do	figureA3.pdf	
Figure A4	code/figures/figure_A4.py	figureA4.pdf	
Figure A5	code/figures/figure_A5.py	figureA5.pdf	
Figure A6	code/figures/figure_A6.do	figureA6a.pdf, figureA6b.pdf	
Figure A7	code/figures/figure_A7.py	figureA7a.pdf, figureA7b.pdf	
Figure A8	code/figures/figure_A8.py	figureA8.pdf	
Figure A9	code/figures/figure_A9.py	figureA9a.pdf, figureA9b.pdf	
Figure A10	code/figures/figure_A10.py	figureA10.pdf	

Figure/Table #	Program	Output file(s)	Note
Figure A11	code/figures/figure_A11.do	figureA11a-d.pdf	needs step 08b
Figure A12	code/figures/figure_A12.do	figureA12.pdf	
Figure A13	code/figures/figure_A13.do	figureA13.pdf	
Figure A14	code/figures/figure_A14.do	figureA14a-d.pdf	
Figure A15	code/figures/figure_A15.do	figureA15a.pdf, figureA15b.pdf	
Figure A16	code/figures/figure_A16.do	figureA16a-d.pdf	
Figure A17	code/figures/figure_A17.do	figureA17a-d.pdf	needs step 07

Expected agreement

Each exhibit **replicates the manuscript exactly**, and all outputs are deterministic across reruns and machines (see "Controlled Randomness" above).

Data Appendix: variable-level documentation

Complete variable lists, extracted directly from the shipped files. PitchBook variable names follow PitchBook's own data dictionary (the platform's field names, lowercased; the .dta files carry the original names as Stata variable labels); PatentsView columns are documented in the PatentsView data dictionary (<https://patentsview.org/download/data-download-dictionary>).

Public author-compiled files

un_polity_china.dta (8 variables) — country–year panel

Variable	Description
year	year
country	ISO 3166 alpha-3 code
polity_score	Polity2 combined score (Polity5)
china_polity_score	China's Polity2 in the same year
un_ideal_point	country's ideal point on the unidimensional scale of UN preferences (Bailey et al. 2017)
china_un_ideal_point	China's ideal point in the same year
un_distance_to_chn	absolute ideal-point distance to China (recomputed in code)
polity_distance_to_chn	absolute Polity2 distance to China (recomputed in code)

OECD_list.dta (4 variables) — one row per OECD member: hqcountry (country name, PitchBook spelling), joinyear (year joined), OECD (=1), OECD_b80s (=1 if member before 1980 — the paper's non-EM definition).

country_panel.dta (12 variables) — one row per country; bilateral measures relative to China (`_cp` suffix), assembled from standard gravity sources (see the paper's Data Appendix for construction):

Variable	Description
<code>hqcountry</code>	country name
<code>fst_distance_dominant_cp</code>	genetic (FST) distance to China, dominant populations
<code>nei_distance_dominant_cp</code>	genetic (Nei) distance to China, dominant populations
<code>lingprox_dominant_cp</code>	linguistic proximity to China
<code>comlang_off_cp</code>	common official language with China (0/1)
<code>freight_cost_air_cp</code>	air-freight cost proxy to China
<code>freight_cost_surface_cp</code>	surface-freight cost proxy to China
<code>distcap_cp</code>	capital-to-capital distance to China
<code>distw_cp</code>	population-weighted distance to China
<code>exports_fob_usd_cp</code>	exports to China (FOB USD)
<code>imports_cif_usd_cp</code>	imports from China (CIF USD)
<code>agree_cp</code>	trade agreement with China (0/1)

EM_policy_constraints.dta (5 variables) — policy × country panel of hand-coded regulatory constraints: `polycname`, `hqcountry`, `loose_same_policy_constraints` (0/1), `same_policy_constraints` (0/1), `policy_constraint_EM_countries` (count of EM countries sharing the constraint).

policy_constrained_sectors.dta (3 variables) — policy → PitchBook subsegment map: `polycname`, `policy_obstacle` (0/1), `subsegment` (the `marketmap` | `segment` | `subsegment` fullname).

critical_subseg.dta (65 variables) — one row per PitchBook subsegment (`fullname_raw`), with 0/1 flags mapping the subsegment to 35 Chinese critical-supply-chain technology items (`CST_*`), 10 "Made in China 2025" priority areas (`MIC2025_*`), and 19 U.S. DoD critical-technology areas (`DoD_*`). Flag names are the (truncated) technology names; the full lists are printed in the paper's appendix.

country_crosswalk.dta (2 variables): `hqcountry` (PitchBook country name) ↔ `country_2digit` (ISO-2 code).

table6_A16 Tradable or not 2nd.xlsx — sheet Sheet1: `subsegment`, `tradable` (Y/N/O tradable-sector classification).

indicators_WDI_MtV_corrected.xlsx (and the three alternative maps) — sheet `Map_to_Variables_no_panel`: `Series Code`, `Series Name`, then one 0/1 column per market-map sector (AgTech, AI ML, Blockchain, ...) assigning WDI indicators to sectors; sheet Sheet1: the indicator-name ↔ series-code list.

CountryDataset.2024019.supplemented.xlsx (confidential folder) – 13 sheets: CountryTable (the compiled country panel used by the exhibits), VC Chart, ScatterPlot, and the cleaning sheets (VC Clean 1969-2000, VC Clean 2001-2021, R&D Clean 1981-2015, R&DClean 1996-2014, R&D Clean 2015-2021, publicationscleaning, patentsto2015cleaning, patents2016to2019cleaning) that document every source series and imputation behind the panel. tableA1_published_values.csv columns: country, emergence_year, gdp_pc_2011usd, vc_pc_2011usd, pct_world_vc, pct_world_pubs, pct_world_rd, pct_us_patents.

table1_panelB_funding_sources.csv columns: source, received_any_pct, share_of_funding_pct (tabulated from Beraja et al. 2024).

Trade_Volume_Export_Import.csv — standard UN Comtrade bulk layout (47 columns: ReporterISO, PartnerISO, FlowCode, RefYear, Fobvalue, Cifvalue, PrimaryValue, ...); the code uses reporter/partner/flow/year and the value columns.

geocoding_resource/ — simplemaps.csv: city, city_ascii, lat, lng, country, iso2, iso3, admin_name, capital, population, id. opendatasoft.csv: the GeoNames "cities with population > 1000" export (semicolon-separated; name / ASCII name / alternate names / coordinates / country / population fields). crosswalk.xlsx: sheet country (country_2digit, country_fullname), sheet US (state, state_2digit). The Natural Earth shapefile sets ship with their original README.html and VERSION.txt.

Confidential files (variable lists)

pbdeals.dta (95 variables) — one row per deal. Key variables used by the pipeline: companyid, companyname, dealid, dealno, dealdate, dealsize (USD m), dealstatus, dealtype/dealtype2/dealtype3, dealclass, premoneyvaluation, postvaluation, raisedtodate, vcround, totalinvestedcapital, investors, newinvestors, followoninvestors, nativecurrencyofdeal, businessstatus, financingstatus, plus IPO fields (tickersymbol, exchange, marketcapendoffirsttradingday, ...), deal-term fields (typeofstock, liquidationpreferences, boardvotingrights, ...), and company financials at deal time (revenue, ebitda, netincome, valuation multiples). All 95 names follow PitchBook's deal-report field names, lowercased, with the original PitchBook names preserved as Stata variable labels inside the file.

pbcompanies.dta (94 variables) — one row per company: identifiers and names (companyid, companyname, companylegalname, ...), the free-text description and keywords used by the ML layers, year-founded, the HQ location block (hqlocation, hqcity, hqstate_province, hqcountry, hqglobalregion, ...), financing summary (firstfinancingdate/size/dealtype ..., lastfinancing*, lastknownvaluation*, totalraised), investor lists (activeinvestors, formerinvestors), and PitchBook's growth/web/social metrics.

pbinvestors.dta (94 variables) — one row per investor: investorid, investorname, description_inv, primaryinvestortype_inv, otherinvestortypes_inv, aum_inv,

yearfounded_inv, the HQ block (hqcountry_inv, ...), activity counts (totalinvestments_inv, totalexits_inv, ...), fund counts, and stated investment preferences (preferred*_inv).

pbdealinvestor.dta (11 variables) — deal-investor links: dealid, investorid, investorname, investorstatus, isleadinvestor, investorfundid, investorfundname, investorwebsite, leadpartnerid, leadpartnername, lastupdated.

LP.dta (51 variables) — one row per fund (4,147 funds): FundID, FundName, Investor, Vintage, CloseDate, FundLocation, FundCity, FundStatus, FundSize, TotalFundInvestments, TotalLPs, then one column per LP country (UnitedStates, China, HongKong, ...; 37 countries + Unknown) counting the fund's limited partners from that country, and Foreign. Input to the Table A9 western-LP analysis (used by step 03c).

company_level_predictions_v2.dta (50 variables) — one row per company (402,695 companies): companyid, hqcountry, then 12 ordered prediction slots fullname1–fullname12 (each a marketmap|segment|subsegment string) with the corresponding marketmap#/segment#/subsegment# splits.

similarity_at_sector_x_country_x_year.dta (25 variables) — sector×country×year: deal counts (first_deal, early, late) and the mean company-level SBERT similarity to Chinese/U.S. firms (china_similarity_mean, _10, _25, us_similarity_*), each also in _afteronly (only pairs where the focal firm is founded after the reference firm) and _placebo variants. similarity_compiled.csv / similarity_compiled_western.csv are the company-level versions (companyid1 × sector with the same measure set; the western file splits Chinese reference firms by western-investor backing).

patent_sector_labels_*.dta — patent→sector label pairs: grantyear (512,162 rows; patent_id, sector, disambig_country, patent_date), new_grantyear (530,765 rows; adds patent_year), new100k (451,328 rows; patent_id, sector only).

patent_countrysector_similarity_2000_2013_100k_new_china.dta (9 variables): country1, country2, sector, and the pairwise patent-text similarity aggregates mean, median, top25, top10, top5, top1. **worldwide_pairs_filtered_no_EPO_countries.dta** (7 variables): sector, country_1, country_2, count (jointly-filed family count), the two country-level totals, and fullname_raw.

Descriptive-exhibit workbooks (confidential, Raw/other_data_resource/): figure1ab_Figures 1a and 1b.xlsx (the authors' direct PitchBook Platform pull of global VC totals by year and region); the three figure1c_IPO_CapitalIQ_to_PB_* workbooks (Capital IQ IPO screens mapped to the paper's EM/US/other-DM groups); figureA4_Comparisons of PB and Other Sources.xlsx and figureA4_Data from Zero2IPO...xlsx (China VC series comparison); table1_panelA_vc_groups.csv (PitchBook VC-group counts); table1_panelC_...xlsx (State Street fund-return summary); table1_panelD_Chinese VC.xlsx (Hurun founder tabulation); tableA1_VCpc.xlsx (PitchBook VC-per-capita tabulation). The hand-corrected/classified items among these — the nonEM_nonUS IPO workbook,

table1_panelA_vc_groups.csv, and tableA1_VCpc.xlsx — sit alongside the rest at the folder root.

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Data and data-adjacent sources (all datasets are also listed in the manuscript's references):

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Acknowledgements

This README follows the Social Science Data Editors template (Vilhuber et al., version 1.2, https://social-science-data-editors.github.io/template_README/).